(64)

Appendix - VI

Item no - 19

DOCTOR OF PHILOSOPHY

IN

COMPUTER SCIENCE AND APPLICATIONS

OFFERED BY

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

PANJAB UNIVERSITY, CHANDIGARH

(Examination 2024 - 2025)

Ph.D. Programme

(65)

Ph.D. Programme

The purpose of the Doctor of Philosophy programme in Computer Science and Applications is to prepare students for research and teaching positions in universities and colleges, and for research and leadership positions in industry and government. The primary aim of the programme is to train students in the methods of scientific inquiry and independent research. This is accomplished through advanced coursework and active participation with the faculty in their research programmes. Doctor of Philosophy students are expected to have a broad knowledge of all fields of computer science and have a deep understanding of at least one of its areas. In addition to this requirement, a Doctor of Philosophy student must be up to date in all the developments in his/her major area of specialization. The most important component of the Doctor of Philosophy programme is learning to perform independent and significant research in one's area of specialization.

Outlines of Tests and Syllabi

Sr. No.	Course No.	Course Title	L	T	P	Credits
1	CSA - 101	*Research Methodology*	3	-	1	4
2	CSA - 102	*Research and Publication Ethics*	4	-	-	4
3	**Elective I:** CSA -103 *Advanced Software Engineering* CSA -104 *Soft Computing Techniques* CSA -105 *Multiagent Decision Making Systems* CSA- 106 *Image Procession and Computer Vision* CSA -107 *Educational Technologies and Advanced Database Systems* *CSA-108 Mobile Communication and Android Application Development* *CSA-109 Parallel and Distributed Computing* *CSA-110 Data Mining and Data Warehousing* *CSA-111 Simulation & Modeling* *CSA-112 Network Technologies* *CSA-113 Machine Learning Techniques*		3	-	1	4

(66)

Examination Scheme

For each paper there will be one midterm examination of 20 marks, seminar presentations of 10 Marks and an end term paper of 70 Marks. The grading scheme for all papers is as under :-

GRADE	MARKS (RANGE)	Numerical grade points
O+	90 – 100%	10
O	80 – 90 (Excluding 90)	9
O-	75 – 80 (Excluding 80)	8
A+	70 – 75 (Excluding 75)	7.5
A	65 – 70 (Excluding 70)	7
A-	60 – 65 (Excluding 65)	6.5
B+	55 – 60 (Excluding 60)	6
F	Less than 55%	0

In case a student is unable to obtain minimum passing grade in a paper but has fulfilled the minimum requirement of attending 75% lectures in that paper, he/she will be given a chance to re-appear in that paper in the immediate next semester.

CGPA (Cumulative Grade Point Average) will be calculated using the formula $\Sigma c_i n_i / \Sigma c_i$

Where c_i = Credits allotted to a paper

n_i = Numerical grade points earned in c_i^{th} paper

The conversion formula is as under :-

CGPA x 10 = Percentage

Chairperson
Department of Computer Science
& Applications
Panjab University
Chandigarh

(67)

Courses of Reading

Research Methodology

Paper Code: **CSA 101**

Objectives: The main objective of the course is to apprise the researchers of the research methodologies, tools and techniques for effective research.

Note :

- The Question paper will consist of four units.
- Examiner will set total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT – I

Introduction: Objectives of Research, Types of Research, Steps in Research, characteristics and Criteria of good research, problems versus research problems, sources of research problems, Research Questions and Hypotheses.

Formulating the Research problem: Writing about the literature reviewed with critical analysis, identifying gaps (if any) in the existing research based on the reviewed literature, considerations in selecting a research problem, steps in formulating a research problem, formulating a research problem in applied, qualitative and quantitative research, proposed research goals and objectives

UNIT – II

Research design: Features of a research design, different research designs, developing a research plan, designs commonly used in qualitative research, random samples; case studies of qualitative, quantitative and applied research problems and their design.

Data Collection and Methods: Techniques for collection of data using primary and secondary sources, methods of data collection in qualitative, quantitative and applied research, standard databases, in-house databases, methods of validity, reliability and verification of output or research findings.

UNIT - III

Processing and Analysis of Data: Role of statistics in research, fundamentals of statistical analysis and inference, correlation and Regression, probability distributions, hypothesis testing, parametric and non parametric tests, Principal Component Analysis, Moments and response curve methods, state vector machines, uncertainty analysis and Bayesian networks; Generalization and interpretation. Role of software tools in preparing and analyzing of data.

UNIT - IV

Research Writing: Structure and components of scientific reports, Importance of effective communication, Bibliography and Referencing, Oral presentation and role of visual aids, use of documentation and presentations tools such as Latex and Microsoft Office, role of PERT chart in research timeline, features of scientific research proposal.

JK

(68)

Suggested Readings:

1. Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, Joseph Bizup and William T. FitzGerald, "The Craft of Research, Fourth Edition", The University of Chicago press, 2016
2. John W. Creswell, "Research DesignQualitative, Quantitative, and Mixed Methods Approaches", FourthEdition, Sage Publishers, 2019
3. C. R. Kothari, "Research Methodology – Methods and techniques", (Second Revised Edition) New Age International Publishers 2009
4. Ranjit Kumar, "Research Methodology: A step by step guide for Beginners", Sage Publications Ltd., 2014
5. Wayne Goddard and Stuart Melville, "Research methodology: An Introduction for science & Engineering students", Juta and Company Ltd., 2004
6. Sreenivasulu N. S., "Law relating Intellectual Property", Partridge Publications, 2013
7. R. pannerselvam, "Research Methodology", PHI, 2005

JK

(69)

Research and Publication Ethics

Paper Code: **CSA 102**

Objectives : This course is a seminar course for Research Scholars in Computer Science and Applications. The aim is to explore literature including recent papers of interest from reputed and refereed Journals, theses and other relevant documents and understand the state of the art research in Computer Science. Each student should identify a base paper related to the proposed research area after making extensive literature review, prepare a report on the paper, submit the document and present the same in the research seminar.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Academic & research writing: Philosophy of Research; Importance of academic writing; Basic rules of academic writing, English in academic writing; Styles of research writing, Plagiarism; Introduction; Tools for the detection of plagiarism; Avoiding plagiarism, Intellectual Honesty and Research Integrity.

UNIT - II

Journal Metrics, Author Metrics, Literature review: Introduction, Source of literature; Process of literature review, Online literature databases; Literature management tools, Review Paper Writing, Research paper writing, Referencing and citation; Research Paper Submission processes.

UNIT - III

Synopsis and Thesis Writing,Empirical research study, Challenges in Indian research & writing; Team management (mentor and collaborators); Time Management, Research proposal writing; Abstract / Conference Paper / Book / Book Chapter writing; OERs: basic concept and licenses.

UNIT - IV

Open Educational Resources (OERs) for learning & Research; OERs development, Ethics in Research: Research fraud, competing interest, authorship, slicing research, FFP, COPE guidelines, Importance of cyber security in research work.

Suggested Readings:

1. Santosh Kumar Yadav, Research and Publication Ethics, Sprnger Cham, 2023, https://doi.org/10.1007/978-3-031-26971-4
2. NimitChowdhary, Sunayana, Monika Prakash, "Research and Publication Ethics An Introduction", Routledge – Taylor and Francis, 2024,
3. Ajay Semalty, Academic Writing, 2021, ISBN: 9789390211739, bs Publications, Hyderabad, India : https://bspublications.net/bookdetail.phd?bid=1574
4. Ajay Semalty, Mona Semalty, Art of Writing and Publishing in Pharmaceutical Journals, 2021, ISBN: 9789386819994, Pjharma Med Press, Hyderabad, India : https://bspublications.net/

JW

Advanced Software Engineering

Paper Code: **CSA 103**

Objectives: This comprehensive course provides a broad perspective on the advanced topics of Software Engineering and embark on more in-depth research or practice in software engineering.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT – I

Software Architecture - Concepts, representation techniques, development methods, and tools for architecture-centric software engineering, decentralized software architecture, architectures for Mobile and Pervasive Systems evaluating software architectures,

Software Configuration Management: Configuration Management, Change Management, Version Management, Build and Release Management.

UNIT - II

Software Reuse: Motivation, Inhibitors, Techniques - Component Based Software Engineering, Process Models, Reuse Metrics.

Software Re-engineering: Introduction, Restructuring and Reverse Engineering, Reengineering existing systems, Data Re-engineering and migration, Reengineering Metrics.

UNIT – III

Software Project Management: Software Project Planning, Conventional Software Management Evolution of Software Economics, Improvement of Software Economics.

Quality: Quality issues in the life-cycle model. Quality planning and management. Aspects of quality: reliability, maintainability, correctness, usability. Risk analysis and management.

UNIT – IV

Measurement: The fundamentals of Measurement Theory. Introduction to Metrics (Quality metrics, cost metrics and process metrics). Function points and object points. Estimation Models and Techniques (Top-down/Bottom Up, Analytical Models (Cocomo, Putnam, etc.), Custom Models.

Testing: Requirements, Unit, integration and system testing, validation. Structural testing: coverage techniques. Behavioral testing: domain testing, finite state testing. Mutation and fault seeding. Tools and instrumentation.

(71)

Suggested Readings:

1.	Mary Shaw and David Garlan.	:	Software Architecture: Perspective on an Emerging Discipline, Prentice –Hall.
2.	Paul Clements, Rick Kazman, and Mark Klein.	:	Evaluating Software Architecture Methods and Case Studies. Addison Wesley.
3.	Len Base, Paul Clements, Rick Kazman.	:	Software Architecture in Practice. Second Edition, Addison Wesley.
4.	I. Sommerville	:	Software Engineering, Addison Wesley.
5.	Walker Royce		Software project management, , Pearson Education.
6.	Stephen T. Albin	:	The Art of Software Architecture: Design methods and Techniques, Wiley
7.	David Budgen	:	Software Design, 2nd edition, Addison-Wesley
8.	William A. Florac and Anita D. Carleton	:	Measuring the Software Process; Addison-Wesley
9.	Fenton, Norman E., Whitty & Iizuka	:	Software Quality Assurance & Measurement: A Worldwide Perspective, Thomson Computer Press
10.	Horch J. W	:	Practical Guide to Software Quality Management, Artech House Computer Science
11.	Kan Stephen H.	:	Metrics and Models in Software Quality Engineering, Addison-Wesley.
12.	Glenford J. Myers	:	The Art of Software Testing, John Wiley & Sons
13.	Robert S. Arnold	:	Software Re-engineering, Robert S. Arnold, IEEE Comp. Society Press.

JK

(72)

Soft Computing Techniques

Paper Code: **CSA 104**

Objectives: The objective of the course is to provide understanding of the soft computing techniques including artificial neural networks, fuzzy logic and genetic algorithms useful for research in intelligent systems and image processing. After going through the course, the candidate will be able to choose and apply the techniques required for the research problem undertaken.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Introduction to Soft Computing: Basic soft computing techniques: Neural networks, Fuzzy logic, Genetic algorithms; Hybrid systems, Application of soft computing.

Artificial Neural Networks: Fundamentals, Characteristics of neural networks, Basic models of artificial neural network: Connections, Learning method, Activation functions; Neural network architectures; Applications of neural networks.

UNIT - II

Supervised Learning Networks : Introduction to supervised learning, Architecture and training algorithms for perception network and back propagation networks.

Unsupervised Learning Networks: Introduction to unsupervised learning, Architecture and training algorithms for Kohonen self-organizing maps and adaptive resonance theory networks (ART1, ART2).

Hopfield Networks: Introduction, Architecture and training algorithm for discrete Hopfield nets.

UNIT - III

Fuzzy Logic: Introduction and Application to Fuzzy logic, Classical sets, Fuzzy Sets, Operations on Fuzzy Sets, Fuzzy Relations, Membership functions, Methods of membership value assignment, and Defuzzification methods, Neuro-fuzzy systems, genetic algorithms and fuzzy logic.

Fuzzy Logic Control Systems: Architecture and Operation of Fuzzy Logic Control (FLC) systems, FLC System Models, Applications of FLC Systems.

UNIT - IV

Introduction to genetic algorithms: Biological background, Genetic algorithms versus traditional algorithms. Basic terminologies in genetic algorithm: genes, fitness and populations. Steps in a general genetic algorithm.

Operations in genetic Algorithm: Selection, Crossover and Mutation, Applications of Genetic Algorithms.

Soft computing Tools: features, applications, and use of tools such as MATLAB, SCILAB, and WEKA, developing small projects using these tools and employing soft computing techniques.

JK

73

Suggested Readings:

1.	S.N. Sivanandam and S.N. Deepa	:	Principles of Soft Computing, Second Edition, John Wiley.
2.	Karray and Silva	:	Soft Computing and Intelligent Systems Design, Pearson Education.
3.	Gallant Stephen	:	Neural Network Learning and Expert System, MIT Press.
4.	S. Rajasekaran and Pai	:	Neural Networks, Fuzzy Logic, and Genetic Algorithms, PHI.
5.	Rao and Rao	:	C++ Neural Networks & Fuzzy Logic, BPB.
6.	Timothy Ross	:	Fuzzy Logic with Engineering Applications, Third Edition, Wily Publications, 2010.
7.	Russel C. Eberhart, Yuhui Shi	:	Computational Intelligence: Concepts to Implementation, Morgan Kaufman, 2007

JK

(74)

Multiagent Decision Making Systems

Paper Code: **CSA 105**

Objectives: The main objective of the course is to provide understanding of distributed artificial intelligence and multiagent systems, applications, agent communication protocols and tools and techniques for developing multiagent decision making systems. After going through the course the candidate will be capable of undertaking research in developing any multiagent decision making system.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Agents and Multiagent Systems : Limitation of artificial intelligence (AI), introduction and advantages of distributed AI, agents, multiagent systems (MAS), applications, agent oriented programming, examples of agents and multiagent systems

Agent Communication : Communication levels, message types, KQML (Knowledge Query and Manipulation Language), KIF (Knowledge Interchange Format), ontologies, agent interaction protocol, co-ordination and co-operation protocols, contract nets, and black board systems.

UNIT - II

Distributed Model for Decision Support : Introduction, agent architecture for distributed decision support system, Case Studies from various areas such as Traffic Management, resource management, ecommerce and disaster management.

Artificial Neural Networks: Introduction to Artificial Neural Networks, Neural Networks Architectures, Supervised and Unsupervised Learning, Perceptron, Back Propagation Networks and Associative Memory.

UNIT - III

Fuzzy Logic: Introduction, Basic Concepts of Fuzzy Logic, Fuzzy Sets, Fuzzy Relations, Fuzzy Graphs, and Fuzzy Arithmetic, Fuzzy If-Then Rules, Fuzzy Implications and Approximate Reasoning, Fuzzy Logic in Control Engineering, Fuzzy Logic and Artificial Intelligence, Fuzzy Logic in Database Management and Information Systems, Fuzzy Logic in Pattern Recognition, Neuro-Fuzzy Systems, Genetic Algorithms and Fuzzy Logic.

Genetic Algorithms: Fundamentals, Encoding, Fitness function, Reproduction, Inheritance Operators, Crossover, Inversion and Deletion, Mutation Operator, and Bit-wise Operators; applications

UNIT - IV

Bayesian Networks: Reasoning under uncertainty, Bayesian Networks, Inference in Bayesian Networks, Decision Networks, and Applications of Bayesian Networks.

Programming using agent development and decision making tools: Features, applications and programming using JADE (Java Agent Development Environment), Java agents, Java agent framework, MATLAB, SCILAB, WEKA, and GENIE, for developing multiagent decision making systems

JW

Suggested Readings:

1. Reza Langari, John Yen : Fuzzy Logic: Intelligence, Control, and Information, 1st Edition, Pearson (Chapters 1-8, 11-13, 16, 17)
2. Laurene V. Fausett : Fundamentals of Neural Networks: Architectures, Algorithms And Applications, 1993, Prentice Hall. (Chapters 1-6)
3. Timothy Ross : Fuzzy Logic with Engineering Applications, 3rd Edition 2010, Wiley
4. Kevin B Korb, Ann Nicholson, Korb B Korb : Bayesian Artificial Intelligence, 2004, Chapman & Hall/CRC. (Chapters 1-5)
5. Russell C. Eberhart, Yuhui Shi : Computational Intelligence: Concepts to Implementation, 1st Edition, 2007. Morgan Kaufman
6. David hand, Heikki Mannila, Padhraic Smyth : Principles of Data Mining, PHI
7. S Rajasekharan, G.A. Vijayalakshmi Pai : Neural Networks, Fuzzy Logic and Genetic Agorithms, PHI
8. Gerhard Weiss : Multiagent Systems edited by, MIT press 1999.
9. Amos Gilat : Amos Gilat, MATLAB: An Introduction with Applications, 3rd Edition, 2008, Wiley

(76)

Image Processing and Computer Vision

Paper Code: **CSA 106**

Objectives: The course enables the students to learn fundamentals of Image processing and computer vision. The candidate after going through the course will be capable of performing research in image processing and computer vision and can develop required algorithms using Matlab and other image processing and computer vision tools.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Introduction to Image Processing: Digital Image Processing, Elements of Image Processing Systems—Image Acquisition, Processing, Communication, Display.

Digital Image Fundamentals:

Uniform and Non-uniform Sampling and Quantization, Basic Relationships between pixels—Neighbours of a pixel, Connectivity, Distance Measures, Imaging Geometry—Perspective transformations, Camera Model, Stereo Imaging.

UNIT - II

Image Transforms: Introduction to Fourier Transform, Discrete Fourier Transform, Properties of the Two – Dimensional Fourier Transform, The Fast Fourier Transform (FFT), Inverse FFT, Walsh, Hadamard and Discrete Cosine Transforms.

Image Enhancement: Histogram Processing, Image Averaging, Smoothing Filters, Sharpening Filters, Low Pass and High Pass Filtering, Generation of Spatial Masks from frequency Domain Specifications, Colour Image Processing—Colours Fundamentals, Colour Models, Pseudo-Colour image processing.

UNIT – III

Image Restoration: Degradation Model, Circulant and Non-circulant Matrices, Algebraic Approach to Restoration, Inverse Filtering, Wiener Filter, Constrained Least Square Restoration, Geometric Transformations.

Image Compression: Fundamentals, Image Compression models and techniques, Low Compression, Image Compressions standards.

Image Segmentation, Representation and Description: Detection of Discontinuities, Edge Linking and Boundary Detection, Hough Transform, Thresholding, Region Oriented Segmentation; Fundamentals of Image Representation and Description

UNIT - IV

Pattern Recognition and Computer Vision: Fundamentals, Applications, feature extraction techniques including Principal Component Analysis (PCA), Local Binary patterns (LBP), Euclidean distance; Classification techniques including Decision Trees, Support Vector Machines, and Artificial Neural Networks; Image Interpretation and Content based Retrieval

Using Image Processing and Computer Vision Tools: Standard and open source tools including MATLAB, SCILAB, OpenCV, Programming using PYTHON, implementation of image processing and computer vision algorithms using these tools

Suggested Readings:

1. Rafael C. Gonzalez, Richard E.Woods, Steven L. Eddins : Digital Image Processing, MGatesmark Publishing, 2009.
2. Jain, Anil K. : Fundamentals of Digital Image Processing, Pearson.
3. Jensen, John R. : Introductory Digital Image Processing, Prentice Hall.
4. Dougherty, Edward R. : Image Processing Digital Techniques.

JK

(77)

Educational Technologies and Advanced Database Systems

Paper Code: CSA 107

Objectives: The main objective of the course is to provide a deep and systematic understanding of the processes involved in conducting research in the area of advanced educational technologies. The course aims to equip the researchers with the fundamental concepts of database design and methodologies required for the research areas in educational technologies.

Note:

- The Question paper will consist of four units.
- The examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT – I

Evolution of Educational Technology: Role of AI and machine learning in personalizing learning experiences, intelligent tutoring systems, cloud-based services for scalable educational platforms, and collaborative tools, Integration of IoT in learning environments.

Instruction, Communication, Media and Learning Designs: Changing trends to virtual/online interaction and Semantic Web; significance, critical perspective and use of design patterns for advanced learning; developing taxonomy for learning designs, learning design representations to document, model and share teaching practices.

UNIT - II

AI and ML for Educational Technologies: Use of Classifications and Clustering techniques in Academic Domain: Logistic Regression, Decision Trees, Random Forest, Support Vector Machine, kNN, Naïve Bayes, K-Means Clustering, Hierarchical Clustering, DBSCAN (Density-Based Spatial Clustering of Applications with Noise), BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies).

Deep Learning Techniques: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs), Generative Adversarial Networks (GANs), Transformers, Self-Supervised Learning, Reinforcement Learning.

UNIT - III

Introduction to Database Systems: Database System Applications, Normalization: 1NF, 2NF, 3NF, BCNF, 4NF and 5NF, SQL Queries, Database Users and Administrators, Transactions and its properties, Serializability: Conflict serializability, View serializability, Testing for serializability, Concurrency Control Techniques: Lock-Based Protocols, Timestamp Based Protocols, Validation-based Protocol.

Recovery Systems and Oracle Concepts: Failure Classification, Storage Structure: Types, Stable Storage Implementation, Data Access, Log-Based Recovery, Advanced Recovery Techniques: Logical Undo Logging, Transaction Rollback, Checkpoints, Remote Backup Systems.

JK

Introduction to SQLPLUS, Referential Integrity, SQLPLUS Reports, Introduction to PL/SQL, Cursors, Triggers, Procedures, Functions, Packages, Large Objects, Creating Users, Remote Data Access.

UNIT - IV

Indexing and Concurrency Control: Types of Single-Level Ordered Indexes, Multilevel Indexes, Dynamic Multilevel Indexes Using B-Trees and B+Trees, Locking Techniques, Concurrency Control and Concurrency Control Techniques Based on Timestamp Ordering.

Distributed Databases: Distributed Data Storage, Network Transparency, Distributed Query Processing, Distributed Transaction Model, Commit Protocols, Coordinator Selection, Deadlock Detection, Recovery and Handling.

Suggested Readings:

1. **Luckin, R., Holmes, W., Griffiths, M., & Forcier, L. B. (2021):** "AI for School Teachers." London: Routledge.
2. **Anderson, T., & Dron, J. (2020):** "Learning Analytics: Measuring the Future of Education." New York: Routledge.
3. **Sharples, M., & Deegan, M. (2019):** "Blockchain in Education." Luxembourg: Publications Office of the European Union.
4. **Siemens, G., & Baker, R. S. J. d. (2018):** "Learning Analytics and Educational Data Mining: Towards Communication and Collaboration." New York: Springer.
5. **Veletsianos, G. (2020):** "Learning Online: The Student Experience." Baltimore: Johns Hopkins University Press.
6. **Weller, M. (2020):** "25 Years of Ed Tech." Athabasca University Press.
7. **Gros, B., & García-Peñalvo, F. J. (2016):** "The Future of Ubiquitous Learning." Springer.
8. **A. Silberschatz, H.F Korth and S. Sudarshan:** Database System Concepts, 3rd ed, 1997, McGraw-Hill International Edition.
9. **R. Elmasri and S.B. Navathe:** Fundamentals of Database Systems, 3rd edition.
10. **C.J. Date:** An Introduction to Database Systems, 7th ed, Addison Wesley, Low Priced Edition, 2000.

(79)

Mobile Communication and Android Application Development

Paper Code: **CSA 108**

Objectives: The course will familiarize the researchers with concepts about mobile communication, its architecture, protocols, and mobile databases. It will also enable them to develop variety of mobile applications using Android.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Mobile communication: Mobile computing architecture, Mobile Devices, Mobile System Networks: Cellular, WLAN, Ad hoc networks; GSM: Services and System Architecture, Radio Interfaces, Localization, Calling, Handover, Security; CDMA: Architecture, Spread Spectrum, Coding Methods; GPRS: Introduction, system Architecture; Introduction to EDGE, LTE.

UNIT - II

Mobile database: Database hoarding techniques, Data Caching, Client Server computing: 2-tier and 3-tier client server architecture; Transactional models, Query processing, Data Recovery process; Classification of Data delivery mechanism: Push based, pull based, Hybrid; Selective tuning and indexing technique, comparative study of various Mobile Operating systems.

UNIT – III

Android Application Development: Android architecture, Android versions and its feature set, Android Market application store, installation of tools and SDK for developing Android applications, Using Android Documentation, Working with Android Emulator.
Building Android Applications: life cycle of an activity, creating User Interface (UI),working with activities, intents, dialogs, fragments etc., applying styles and themes to activities, displaying pictures and menus with views, managing application resources, configuring the Android Manifest file, registering activities, designating the launch activity, managing application permissions.

UNIT - IV

Android Data Persistence: saving simple data, writing and reading files in internal and external storage, Creating and using a SQLite database, building data base applications.
Android Networking: obtaining geographical data using GPS, Cell-ID, and Wi-Fi triangulation, monitoring for a location, building a Location Tracker application, connecting to web, using JSON web services: Publishing Android Applications.

JK

Suggested Readings:

1.	Kamal, Raj	:	Mobile Computing, Oxford Higher Education.
2.	Jochen Schiller	:	"Mobile Communications", PHI/Pearson Education, Second Edition.
3.	Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober	:	"Principles of Mobile Computing", Springer, New York, 2003.
4.	Reto Meier	:	Professional Android 2 Application Development, Wrox Publications.
5.	Wei-Meng Lee	:	Beginning Android Application Development, Wiley Publications.
6.	Shane Conder, Lauren Darcey	:	Sams Teach Yourself Android™ Application Development in 24 Hours.

Parallel and Distributed Computing

Paper Code: **CSA 109**

Objectives: The course enables students to learn details of parallel and distributed computing. After completing the course, the candidate will be capable of performing research in parallel and distributed computing.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT – I

Parallel Computing: Taxonomy of parallel processing. Parallel techniques and paradigms; Abstract models of parallel machines; Strategies and paradigms for parallel algorithm design; Correlations of algorithmic to architectural spaces;

UNIT – II

Design methodologies for parallel algorithms for parallel computing platforms; Designing parallel algorithms for various application areas: linear algebra, matrix multiplication, sorting, PDE, graphs, combinatorial search, document classification;

Design methodologies for parallel genetic algorithms parallel programming with MPI, Parallel programming with OpenMP, Hybrid parallel programming MPI+OpenMP.
Multithreaded programming, Parallel program benchmarking, Parallelism profiling

UNIT – III

Cluster Computing: The Role of Clusters, Architecture and Cluster Software, Network Hardware, Network Software, Protocols, Distributed File Systems, Virtualization technologies, Benchmarks

Grid Computing: Role of a Grid, infrastructure, Grid Architecture, Overview of Resource Managers, Application Management; Grid Application Description Languages, Application Partitioning, Meta-scheduling, Mapping ,Monitoring, Web Services, Grid Portals.

UNIT – IV

Cloud Computing: Fundamentals, private, Public and hybrid clouds, characteristics, examples, and applications, difference between cloud computing and distributed computing

Suggested Readings:

1. Grama A., Gupta A., Karypis G., Kumar V : Introduction to Parallel Computing, PEARSON, Addison Wesley
2. Barry Wilkinson, Michael Allen : Parallel Programming Techniques and Applications Using Networked Workstations and Parallel Computers Pearson Prentice Hall .
3. Haupt R., S. Haupt : Practical Genetic Algorithms, Wiley Interscience, A John Wiley & Sons
4. M. Quinn : Parallel Programming in C with MPI and OpenMP, McGraw Hill
5. Hwang K., Zhiwei X. : Scalable Parallel Computing: Technology, Architecture, Programming, McGraw-Hill.
6. Timothy Chou : Fundamentals of Cloud Computing, Cloudbook, Inc., 2015

(82)

Data Mining and Data Warehousing

Paper Code: **CSA 110**

Objectives : This course will introduce concepts and techniques of data mining and data warehousing, including concept, principle, architecture, design, implementation, application of data warehousing and data mining. Some systems for data warehousing and/or data mining will also be introduced to study the methodology of engineering legacy databases for data warehousing and data mining to derive business rules for decision support systems.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Introduction : Data Warehousing, Characteristics of a Data Warehouse.

Developing Data Warehouse: Building a Data Warehouse, Data Warehouse architectural strategies, Design considerations, metadata, Tools for Data Warehousing, performance & Technological considerations, Crucial decisions in Designing a Data Warehouse.

UNIT - II

Security : Security in Data Warehouses, Difficulties in implementation of Data Warehouses, Open issues in data Warehouses.

Data Marts: Data Marts, Types of Data Marts, Developing Data Mart based Data Warehouses.

UNIT - III

OLTP and OLAP Systems: Data Modeling, Star Schema for multidimensional view, Snowflake schema

OLAP tools, Managed Query Environment. OLTP versus OLAP.

UNIT - IV

Data Mining : Introduction, Steps of Data Mining Process, Types of Data Mining Tasks,

Data Mining Algorithms : Association (Aprior Algorithm) a nd Classification (Decision Tree Induction),

Back propagation Neural Network, Clustering, Tools for Data Mining.

Programming Using Standard & Open Source Data Warehousing and Data Mining Tools like ORACLE Data Mining, Hdoop and MATLAB.

Suggested Readings:

1. Cornel and Rao : "Database Principles" Cengage, Publications.
2. Benson Alex and Smith Stephen J. : Data Warehousing, Data Mining and OLAP,TMH.
3. Jiawei Han, Micheline Kamber, 2000 : Data Mining : Concepts and Techniques, Morgan Koffman Elscvier.
4. Inmon, W.H., 1999 : Managing the Data Warehouse, C. L. Gassey, John Wiley.
5. Pujari, Arun K. : Data Mining Techniques, University Press.
6. Mattison, 1999 : Data Warehousing and Knowledge Management, Tata McGraw Hill.
7. Fayyad, Usama M.,1996 : Advances in Knowledge Discovery and Data Mining, MIT Press.
8. Elmasi & Navathe : "Fundamentals of Database Systems" Pearson Education.

JW

(83)

Simulation & Modeling

Paper Code: **CSA 111**

Objectives: The course aim to apprise the researchers with simulation and modeling techniques for effective research. After going through the course, the candidate will be able to choose and apply these techniques for their research areas.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Introduction to Simulation: Simulation: Concepts, Areas of Application, Systems and System Environment, Components of a system, Discrete and Continuous Systems, Model of a System, Types of Models, Discrete-Event System Simulation, Steps in a Simulation Study.

System Studies: Subsystems, A Corporate Model, Environment Segment, Production Segment, Management Segment, The Full Corporate Model, Types of System Study, System Analysis, System Design, System Postulation.

UNIT - II

System Simulation: The Technique of Simulation, The Monte Carlo Method, Comparison of Simulation and Analytical Methods, Experimental Nature of Simulation, Types of System Simulation, Numerical Computation Technique for Continuous Models, Distributed Lag Models, Cobweb Models.

System Dynamics: Exponential Growth Models, Exponential Decay Models, Modified Exponential Growth Models, Logistic Curves, System Dynamics Diagrams, Simple System Dynamics Diagrams, Multi-Segment Models, Representation of Time Delays.

UNIT – III

Probability Concepts in Simulation: Stochastic Variables, Discrete Probability Functions, Continuous Probability Functions, Measures of Probability Functions, Numerical Evaluation of Continuous Probability Functions, Continuous Uniformly Distributed Random Numbers, Computer Generation of Random Numbers, A Uniform Random Number Generator, Generating Discrete Distributions, Non-Uniform Continuously Distributed Random Numbers, The Rejection Method.

Arrival Patterns and Service Times: Congestion in Systems, Arrival Patterns, Poisson Arrival Patterns, The Exponential Distribution, The Coefficient of Variation, The Erlang Distribution, The Hyper-Exponential Distribution, Service Times, The Normal Distribution, Queuing Disciplines, Queuing notation, Measures of Queues, Mathematical Solutions of Queuing Problems.

UNIT - IV

Discrete System Simulation: Discrete Events, Representation of Time, Generation of Arrival Patterns, Case Study of System Simulations, Simulation Programming Tasks, Gathering Statistics, Counters and Summary Statistics, Measuring Utilization and Occupancy, Recording Distributions and Transit Times, Discrete Simulation Languages.

JK

(84)

Input Modelling: Data Collection, Identifying the Distribution with Data, Parameter Estimation, Selecting Input Models without Data.

Visualization: Fundamentals, techniques and applications.

Finite Element Analysis: Finite Element method and practical application to different problems

Simulation Software: Simulation in C++,Other open source simulations tools like NS3

Suggested Readings:

1. Averil,M.Law, W. David Kelton : Simulation Modeling and Analysis, 3/e,Mc Graw Hill Publications 2000.
2. C.S. Krishnamoorthy : Finite Element Analysis: Theory and Programming.
3. Jerry Banks, "Handbook of Simulation : Principles, Methodology, Advances, Applications, and Practice", John Wiley & Sons, Inc., 1998.
4. George S. Fishman : "Discrete-Event Simulation: Modeling, Programming and Analysis", Springer-Verlag New York, Inc., 2001.
5. Andrew F. Seila, Vlatko Ceric, Pandu Tadikamalla : "Applied Simulation Modeling", Thomson Learning Inc., 2003.

JK

Network Technologies

Paper Code: **CSA 112**

Objectives: This challenging and comprehensive course provides a broad perspective on the networks and related technologies.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

UNIT - I

Overview of wireless networks: Introduction to wireless communication, Cellular Networks and generations, IEEE 802.11, IEEE 802.16 and IEEE 802.20.

IPv6: Basic protocol, extensions and options, support for Quality of Service (QoS), security, neighbor discovery, auto-configuration, routing.

UNIT - II

Traffic Management: Economic Framework, Traffic Models, Traffic Classes, Scheduling, Admission Control, Peak Load Pricing.

QoS and Security issues: Network security requirements, issues and challenges in security and QoS provisioning, classifications of QoS solutions, IP security overview, IP Security Architecture.

UNIT – III

Mobility in networks: Mobile IP and related issues like Route Optimization, Handoff and Security.

Mobile Ad hoc Networks (MANETs): Introduction to Ad hoc wireless networks and sensor networks, Various MANET issues.

UNIT – IV

Transport layer: Various transport layer solutions and TCP over Ad hoc wireless networks, TCP extensions for high-speed networks, other new options in TCP, TCP in Wireless Domain.

Network Simulation: Elements of queuing model, role of exponential distribution, Event graphs of queuing model, Discrete-event and Continuous Simulation, Introduction to NS3 and Qualnet.

Suggested Readings:

1. William Stallings : Wireless Communication and Networks, Pearson
2. C. Siva Ram Murthy and B. S Manoj : Adhoc Wireless Networks-Architecture and Protocols, Pearson Education
3. W. R. Stevens : TCP/IP Illustrated, Vol. 1, Pearson Education
4. M. Gonsalves and K. Niles : IPv6 Networks, McGraw Hill
5. S. Keshav : An Engineering Approach to Computer Networking, Pearson Education
6. Kamal, Raj : Mobile Computing, Oxford Higher Education.
7. Jochen Schiller : Mobile Communications, PHI/Pearson Education, Second Edition.
8. Requests for Comments (RFCs) & Internet Drafts : Published by Internet Engineering Task Force (www.rfc-editor.org)
9. Journals: IEEE Journal on Selected Areas in Communications : IEEE Transactions on Communication, ACM/IEEE Transactions on Networking.

(86)

Machine Learning Techniques

Paper Code: **CSA 113**

Objectives: The goal of this course is to provide PhD students with an understanding of basic concepts of Machine Learning techniques.

Note :

- The Question paper will consist of four units.
- Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi.
- The students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks.

Course Outcomes :

Students will have the knowledge and skills to:

- Demonstrate knowledge of Machine Learning fundamentals and their role in real-life applications.
- Implement Machine Learning based problems for medium and large scale datasets.
- Data Normalization, Regression, Classification, Clustering, Dimension reduction and Deep Learning.

UNIT I

1. Introduction to Machine Learning

Motivation and role of machine learning in computer science and problem solving, History and importance of machine learning, Role of data, Connection to the knowledge /experience in learning.

2. Machine Learning Essentials

Programming framework for machine learning, essentials of Mathematics in machine learning, role of data visualization and feature extraction machine learning, recent tools in machine learning.

UNIT II

3. Data Preprocessing and Regression

Understanding of data and its preprocessing, normalizing data, regression role in machine learning, multiple and polynomial regression, support vector regression, decision tree regression and random forest regression.

4. Classification

Classification in machine Learning, Logistic classifier, Naive Bayes classification, k nearest neighbor classifier, neural network classification, support vector classification, decision tree classification and random forest classification.

JK

UNIT III

5. Clustering and Data Reduction

Role of clustering and data reduction in machine learning, k means and hierarchical clustering, principal component analysis and linear discriminant analysis.

6. Natural Language Processing

Natural Language Processing - Problems and perspectives, Computational Phonetics and Speech Processing, Tokenisation and Sentence splitting, Computational Morphology, Computational Syntax, Computational Semantics, Introduction of large language models.

UNIT IV

7. Deep Learning - I

Deep Learning requirements and implementation, common deep learning techniques as convolutional neural networks and recurrent neural networks.

8. Deep Learning - II

Generative Adversarial Network, LSTM, embeddings, word-vectors, transformers and attention based networks, transfer learning, autoencoders, Deep Reinforcement Learning.

Suggested Readings:

1. YoshuaBengio, Ian Goodfellow, Aaron Courville, "Deep Learning (Adaptive Computation and ML Series)", MIT Press, 2016.
2. EthemAlpaydin, ""Introduction to Machine Learning", MIT Press, 2020.
3. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.
4. Andreas C. Müller, Sarah Guido, "Introduction to Machine Learning with Python", First Edition, O'Reilly Media, Inc., 2016.
5. AurélienGéron, "Hands-on ML with Scikit-Learn, Keras&TensorFlow", Second Edition, O'Reilly Media, Inc., 2017.
6. Christopher Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
7. Sebastian Raschka and VahidMirjalili, "Python Machine Learning", Packt, 2017.